LinkedIn edunet foundation

AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

[STUDENT ACADEMIC PERFORMANCE ANALYSIS AND GRADE]

Project Domain

[Excel]

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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[Rishi UBR Women's College]

Year - 2026

ACKNOWLEDGEMENT

We would like to take this opportunity to express our deep sense of gratitude to all individuals who helped us directly or indirectly during this project work. Firstly, we would like to thank our supervisor, Umamaheshwari, for being a great mentor and the best adviser we could have asked for. We extend our sincere thanks to the Edunet Foundation, Microsoft, LinkedIn, and SAP for providing this fellowship opportunity to develop and showcase our AI skills. We are also grateful to our institution, Rishi UBR Women's College, for providing the necessary infrastructure.

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ABSTRACT OF THE PROJECT

This project focuses on the "Student Academic Performance Analysis and Grade" system, developed as part of the AICW Fellowship Program. The project addresses the challenge educational institutions face in tracking student progress over multiple semesters without a structured system. Using Microsoft Excel, a dataset of 561 student records was analyzed across six semesters to calculate average marks, grades, and pass/fail status. The methodology involved data cleaning to handle missing values and duplicates, followed by the generation of key performance indicators (KPIs) such as highest/lowest scorers and attendance correlation. The results provide data-driven insights to identify at-risk students and improve academic planning.

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CHAPTER 1

Introduction

This chapter introduces the project undertaken as part of the AICW Fellowship Program. It outlines the background, the problem being addressed, the motivation behind this work, and the objectives and scope of the project. The fellowship projects span multiple domains including Data Analytics, Business Intelligence, Healthcare Analytics, Content Generation, and AI-assisted Reporting.

1.1 Problem Statement

Educational institutions struggle to obtain a clear overview of student progress over time because tracking semester-wise improvement manually is time-consuming and prone to inconsistency. Without a structured analysis system, students with declining performance are not identified at an early stage, and academic planning is often based on assumptions rather than data.

1. Institutions struggle to obtain a clear overview of student progress over time.

2. Students with declining or consistently low performance are not identified at an early stage.

3. Tracking semester-wise improvement or decline manually is time-consuming and prone to inconsistency

4. Academic planning and improvements are often based on assumptions rather than analyzed data.

5. Students do not receive structured performance insights that can guide their improvement.

6. Several factors influence academic performance, including: Socio-economic background Learning environments Teaching methodologies Motivation levels Access to academic support

7. Institutions often face challenges due to insufficient data-driven interventions.

8. These issues make it difficult for institutions to address academic problems effectively.

9. There is a need to better understand the underlying factors affecting student performance.

10. Targeted strategies must be developed to improve overall academic outcomes.

1.2 Motivation

The motivation behind this project is to support data-driven decision-making for academic planning and to enable early intervention for struggling students by correlating attendance with marks.

1.3 Objectives

To maintain organized academic records of students across six semesters.

- To calculate overall performance indicators, including average marks and grades.
- To classify students based on predefined grading standards (e.g., A, B, C, F).
- To identify students who require academic support through the analysis of "Pass/Fail" rates and attendance.

1.4 Scope of the Project

The project covers the storage and analysis of academic data for 561 students across 12 columns, including Semester 1–6 marks, average marks, grade, status, and attendance percentage.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

A review of relevant literature or previous work, identifying gaps that the current project addresses.

2.2 Tools and Technologies Review

- **Microsoft Excel:** Used for "Data Ingestion" and "Data Cleaning" of the 561-row dataset.

- **Microsoft Copilot:** Acted as an AI assistant to generate formulas and summarize complex student trends.

2.3 Gaps Identified

A summary of limitations in current solutions, such as the lack of tool-based solutions for non-technical users or limited documentation on integrating AI with domain-specific analysis.

CHAPTER 3

Proposed Methodology

This chapter describes the methodology adopted to achieve the project objectives. It covers the system design, modules used, data flow, advantages, and requirement specifications.

3.1 System Design

The system is designed as a centralized academic data management tool using Excel. It processes raw student data to generate insights such as course-wise average marks and grade distribution.

3.2 Modules / Components Used

1.Data Cleaning

- Checked for missing values and handled them appropriately.
- Removed duplicate records if any.
- Standardized column names and data formats.
- Ensured numerical columns are in correct datatype.

2.Business KPI

Key Performance Indicators (KPIs) are measurable values used to evaluate student academic performance and overall progress, Like the effectiveness of teaching , subject wise curriculum difficulty analysis, academic success rate, learning progress of student, performance analysis and what should be improved.

3.KPI List

- Average Marks per Semester
- Overall Percentage
- Pass/Fail Rate
- Highest Scoring Student
- Lowest Scoring Student
- Highest Attendance%
- Lowest Attendance%

- High Distinction

3.3 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

3.4 Advantages

- **Automated Data Management**: The system automates repetitive data cleaning tasks, such as removing duplicates and standardizing formats, which saves significant manual effort.

- **Visual Clarity**: It provides clear, interactive visual insights through dashboards that are easy to understand for non-technical stakeholders like faculty and parents.

- **Early Intervention**: By correlating attendance with academic marks, the system enables teachers to identify struggling students at an early stage for remedial support.

- **Enhanced Accuracy with AI**: Leveraging AI tools like Microsoft Copilot accelerates data analysis and reduces human error in complex calculations.

- **Scalability**: The methodology developed for these 561 records can be easily applied to larger datasets or adapted for different academic departments.

- **Cost-Effectiveness**: The project uses widely available tools like Microsoft Excel and Power BI, making it an affordable solution within the existing Microsoft 365 ecosystem.

- **Data-Driven Planning**: It replaces assumptions with analyzed data, allowing management to make better decisions regarding curriculum improvements and teaching strategies.

- **Standardized Evaluation**: The system ensures a consistent and fair grading process across all students based on predefined academic standards.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4

Implementation and Results

This chapter presents the actual implementation of the project and the results obtained at each stage. Screenshots of dashboards, cleaned datasets, or generated content should be inserted where indicated.

4.1 Data Cleaning Results

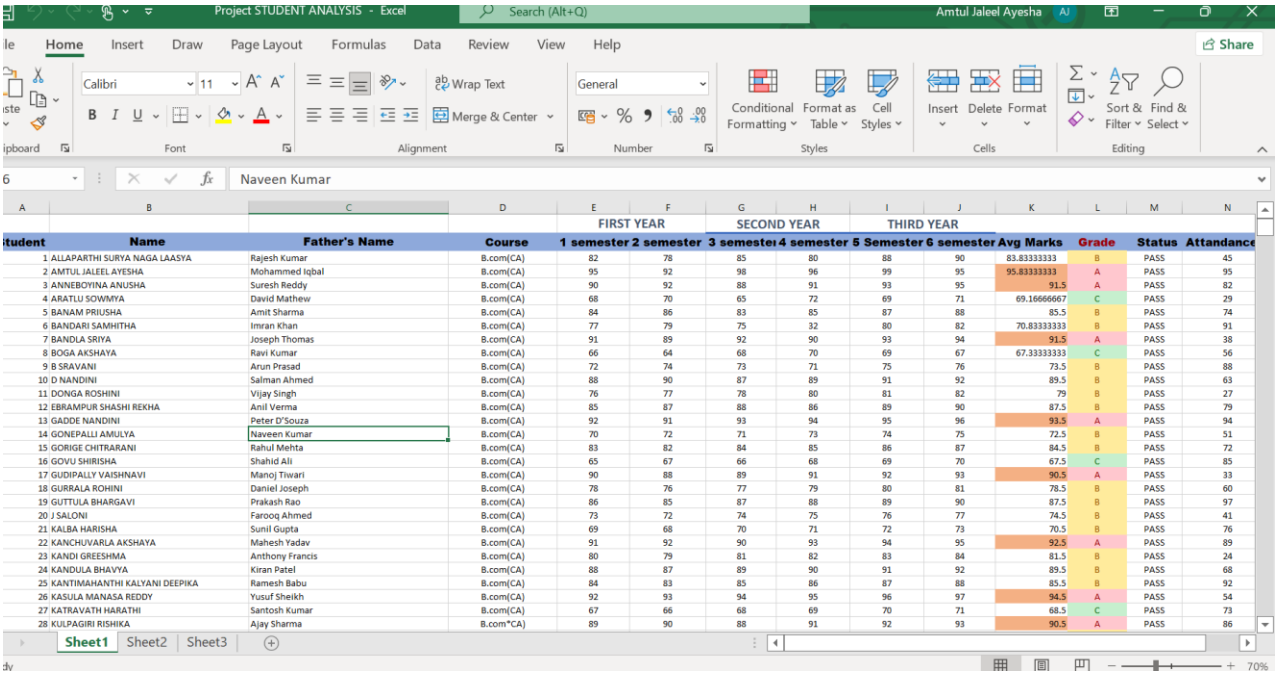
The data cleaning process ensured that all numerical columns were in the correct data type and standardized column names for 561 records.

• Checked for missing values and handled them appropriately.

• Removed duplicate records if any.

• Standardized column names and data formats.

• Ensured numerical columns are in correct datatype.



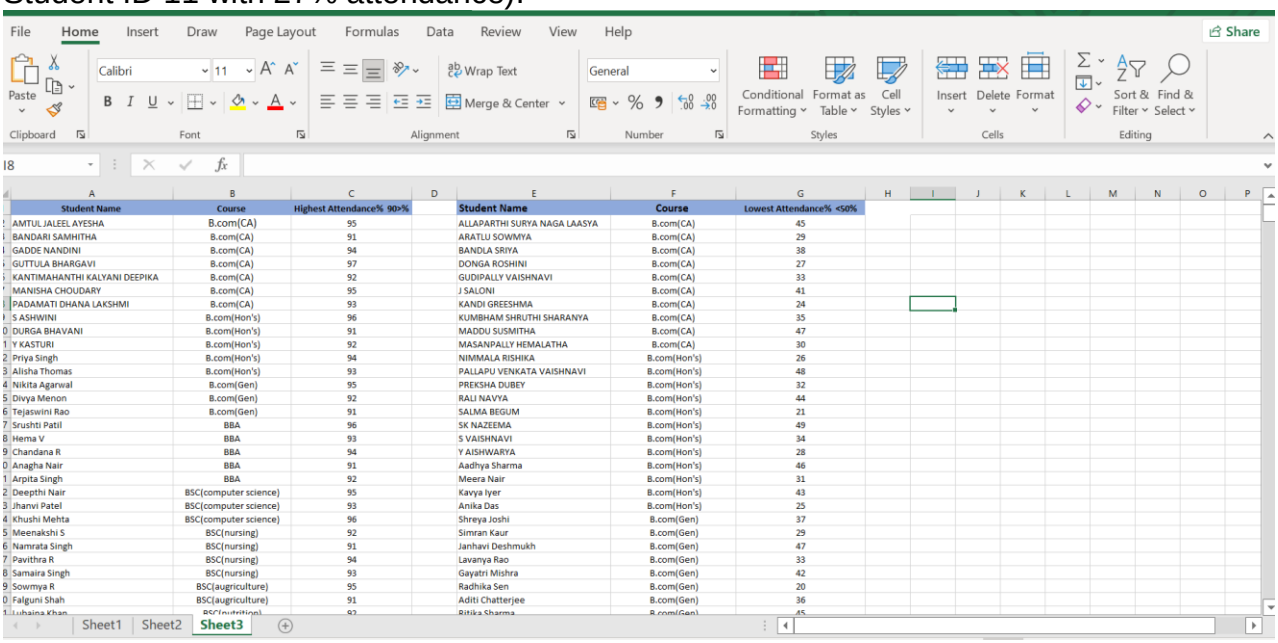
				FIRST YEAR		SECOND YEAR		THIRD YEAR					
tudent	Name	Father's Name	Course	1 semester	2 semester	3 semester	4 semester	5 Semester	6 semester	Avg Marks	Grade	Status	Attandance
1	ALLAPARTHI SURYA NAGA LAASYA	Rajesh Kumar	B.com(CA)	82	78	85	80	88	90	83.83333333	B	PASS	45
2	AMTUL JALEEL AYESHA	Mohammed Iqbal	B.com(CA)	95	92	98	96	99	95	95.83333333	A	PASS	95
3	ANNEBOYINA ANUSHA	Suresh Reddy	B.com(CA)	90	92	88	91	93	95	91.5	A	PASS	82
4	ARATLU SOWMYA	David Mathew	B.com(CA)	68	70	65	72	69	71	69.16666667	C	PASS	29
5	BANAM PRIUSHA	Amit Sharma	B.com(CA)	84	86	83	85	87	88	85.5	B	PASS	74
6	BANDARI SAMHITHA	Imran Khan	B.com(CA)	77	79	75	32	80	82	70.83333333	B	PASS	91
7	BANDLA SRIYA	Joseph Thomas	B.com(CA)	91	89	92	90	93	94	91.5	A	PASS	38
8	BOGA AKSHAYA	Ravi Kumar	B.com(CA)	66	64	68	70	69	67	67.33333333	C	PASS	56
9	B SRAVANI	Arun Prasad	B.com(CA)	72	74	73	71	75	76	73.5	B	PASS	88
10	D NANDINI	Salman Ahmed	B.com(CA)	88	90	87	89	91	92	89.5	B	PASS	63
11	DONGA ROSHINI	Vijay Singh	B.com(CA)	76	77	78	80	81	82	79	B	PASS	27
12	EBRAMPUR SHASHI REKHA	Anil Verma	B.com(CA)	85	87	88	86	89	90	87.5	B	PASS	79
13	GADDE NANDINI	Peter D'Souza	B.com(CA)	92	91	93	94	95	96	93.5	A	PASS	94
14	GONEPALLI AMULYA	Naveen Kumar	B.com(CA)	70	72	71	73	74	75	72.5	B	PASS	51
15	GORIGE CHITRARANI	Rahul Mehta	B.com(CA)	83	82	84	85	86	87	84.5	B	PASS	72
16	GOVU SHIRISHA	Shahid Ali	B.com(CA)	65	67	66	68	69	70	67.5	C	PASS	85
17	GUDIPALLY VAISHNAVI	Manoj Tiwari	B.com(CA)	90	88	89	91	92	93	90.5	A	PASS	33
18	GURRALA ROHINI	Daniel Joseph	B.com(CA)	78	76	77	79	80	81	78.5	B	PASS	60
19	GUTTULA BHARGAVI	Prakash Rao	B.com(CA)	86	85	87	88	89	90	87.5	B	PASS	97
20	J SALONI	Farooq Ahmed	B.com(CA)	73	72	74	75	76	77	74.5	B	PASS	41
21	KALBA HARISHA	Sunil Gupta	B.com(CA)	69	68	70	71	72	73	70.5	B	PASS	76
22	KANCHUVARLA AKSHAYA	Mahesh Yadav	B.com(CA)	91	92	90	93	94	95	92.5	A	PASS	89
23	KANDI GREESHMA	Anthony Francis	B.com(CA)	80	79	81	82	83	84	81.5	B	PASS	24
24	KANDULA BHAVYA	Kiran Patel	B.com(CA)	88	87	89	90	91	92	89.5	B	PASS	68
25	KANTIMAHANTHI KALYANI DEEPIKA	Ramesh Babu	B.com(CA)	84	83	85	86	87	88	85.5	B	PASS	92
26	KASULA MANASA REDDY	Yusuf Sheikh	B.com(CA)	92	93	94	95	96	97	94.5	A	PASS	54
27	KATRAVATH HARATHI	Santosh Kumar	B.com(CA)	67	66	68	69	70	71	68.5	C	PASS	73
28	KULPAGIRI RISHIKA	Ajay Sharma	B.com*CA)	89	90	88	91	92	93	90.5	A	PASS	86

4.2 Visualization / Output Results

Average Marks: Calculated for each student (e.g., Student ID 2 achieved an average of 95.83).

- **Grade Distribution:** Students were assigned grades such as 'A' for high performance and 'F' for failing (e.g., Student ID 69 failed with an average of 49.8).

- **Attendance Tracking:** The system identified students with low attendance (e.g., Student ID 11 with 27% attendance).



Student Name	Course	Highest Attendance% 90>%
AMTUL JALEEL AYESHA	B.com(CA)	95
BANDARI SAMHITHA	B.com(CA)	91
GADDE NANDINI	B.com(CA)	94
GUTTULA BHARGAVI	B.com(CA)	97
KANTIMAHANTHI KALYANI DEEPIKA	B.com(CA)	92
MANISHA CHOUDARY	B.com(CA)	95
PADAMATI DHANA LAKSHMI	B.com(CA)	93
S ASHWINI	B.com(Hon's)	96
DURGA BHAVANI	B.com(Hon's)	91
Y KASTURI	B.com(Hon's)	92
Priya Singh	B.com(Hon's)	94
Alisha Thomas	B.com(Hon's)	93
Nikita Agarwal	B.com(Gen)	95
Divya Menon	B.com(Gen)	92
Tejaswini Rao	B.com(Gen)	91
Srushti Patil	BBA	96
Hema V	BBA	93
Chandana R	BBA	94
Anagha Nair	BBA	91
Arpita Singh	BBA	92
Deepthi Nair	BSC(computer science)	95
Jhanvi Patel	BSC(computer science)	93
Khushi Mehta	BSC(computer science)	96
Meenakshi S	BSC(nursing)	92
Namrata Singh	BSC(nursing)	91
Pavithra R	BSC(nursing)	94
Samaira Singh	BSC(nursing)	93
Sowmya R	BSC(augriculture)	95
Falguni Shah	BSC(augriculture)	91

Student Name	Course	Lowest Attendance% <50%
ALLAPARTHI SURYA NAGA LAASYA	B.com(CA)	45
ARATLU SOWMYA	B.com(CA)	29
BANDLA SRIYA	B.com(CA)	38
DONGA ROSHINI	B.com(CA)	27
GUDIPALLY VAISHNAVI	B.com(CA)	33
J SALONI	B.com(CA)	41
KANDI GREESHMA	B.com(CA)	24
KUMBHAM SHRUTHI SHARANYA	B.com(CA)	35
MADDU SUSMITHA	B.com(CA)	47
MASANPALLY HEMALATHA	B.com(CA)	30
NIMMALA RISHIKA	B.com(Hon's)	26
PALLAPU VENKATA VAISHNAVI	B.com(Hon's)	48
PREKSHA DUBEY	B.com(Hon's)	32
RALI NAVYA	B.com(Hon's)	44
SALMA BEGUM	B.com(Hon's)	21
SK NAZEEMA	B.com(Hon's)	49
S VAISHNAVI	B.com(Hon's)	34
Y AISHWARYA	B.com(Hon's)	28
Aadhya Sharma	B.com(Hon's)	46
Meera Nair	B.com(Hon's)	31
Kavya Iyer	B.com(Hon's)	43
Anika Das	B.com(Hon's)	25
Shreya Joshi	B.com(Gen)	37
Simran Kaur	B.com(Gen)	29
Janhavi Deshmukh	B.com(Gen)	47
Lavanya Rao	B.com(Gen)	33
Gayatri Mishra	B.com(Gen)	42
Radhika Sen	B.com(Gen)	20
Aditi Chatterjee	B.com(Gen)	36

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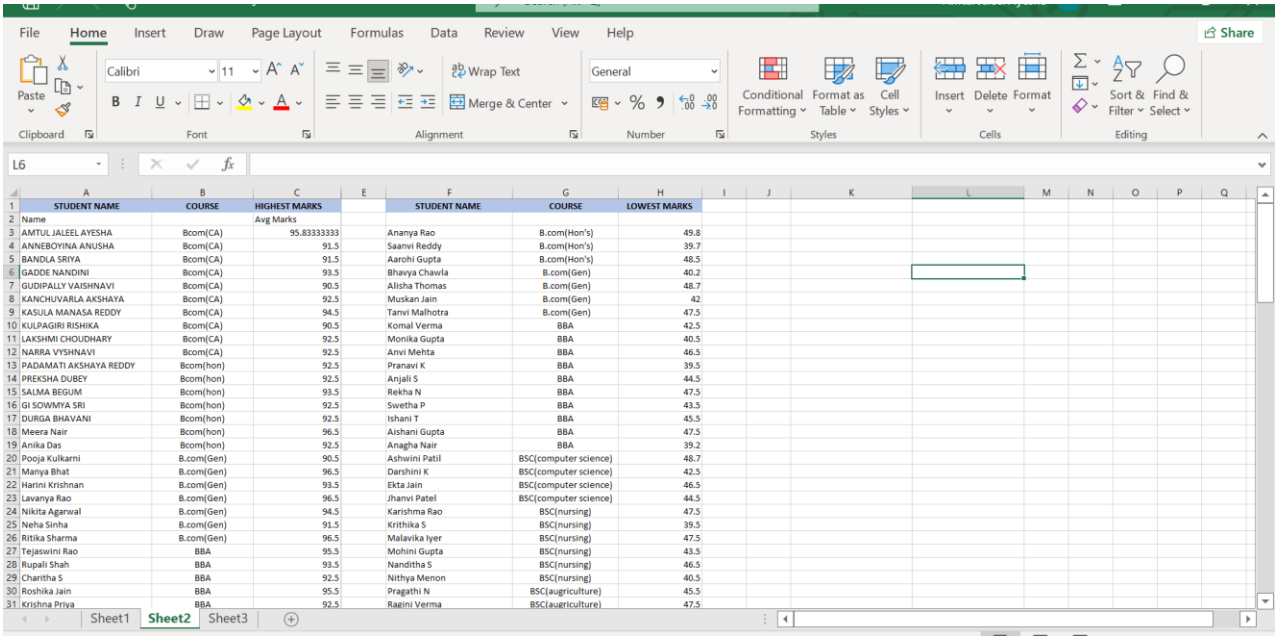

STUDENT NAME	COURSE	HIGHEST MARKS
Name		Avg Marks
AMTUL JALEEL AYESHA	Bcom(CA)	95.83333333
ANNEBOYINA ANUSHA	Bcom(CA)	91.5
BANDLA SRIYA	Bcom(CA)	91.5
GADDE NANDINI	Bcom(CA)	93.5
GUDIPALLY VAISHNAVI	Bcom(CA)	90.5
KANCHUVARLA AKSHAYA	Bcom(CA)	92.5
KASULA MANASA REDDY	Bcom(CA)	94.5
KULPAGIRI RISHIKA	Bcom(CA)	90.5
LAKSHMI CHOUDHARY	Bcom(CA)	92.5
NARRA VYSHNAVI	Bcom(CA)	92.5
PADAMATI AKSHAYA REDDY	Bcom(hon)	92.5
PREKSHA DUBEY	Bcom(hon)	92.5
SALMA BEGUM	Bcom(hon)	93.5
GI SOWMYA SRI	Bcom(hon)	92.5
DURGA BHAVANI	Bcom(hon)	92.5
Meera Nair	Bcom(hon)	96.5
Anika Das	Bcom(hon)	92.5
Pooja Kulkarni	B.com(Gen)	90.5
Manya Bhat	B.com(Gen)	96.5
Harini Krishnan	B.com(Gen)	93.5
Lavanya Rao	B.com(Gen)	96.5
Nikita Agarwal	B.com(Gen)	94.5
Neha Sinha	B.com(Gen)	91.5
Ritika Sharma	B.com(Gen)	96.5
Tejaswini Rao	BBA	95.5
Rupali Shah	BBA	93.5
Charitha S	BBA	92.5
Roshika Jain	BBA	95.5
Krishna Priya	BBA	92.5

STUDENT NAME	COURSE	LOWEST MARKS
Ananya Rao	B.com(Hon's)	49.8
Saanvi Reddy	B.com(Hon's)	39.7
Aarohi Gupta	B.com(Hon's)	48.5
Bhavya Chawla	B.com(Gen)	40.2
Alisha Thomas	B.com(Gen)	48.7
Muskan Jain	B.com(Gen)	42
Tanvi Malhotra	B.com(Gen)	47.5
Komal Verma	BBA	42.5
Monika Gupta	BBA	40.5
Anvi Mehta	BBA	46.5
Pranavi K	BBA	39.5
Anjali S	BBA	44.5
Rekha N	BBA	47.5
Swetha P	BBA	43.5
Ishani T	BBA	45.5
Aishani Gupta	BBA	47.5
Anagha Nair	BBA	39.2
Ashwini Patil	BSC(computer science)	48.7
Darshini K	BSC(computer science)	42.5
Ekta Jain	BSC(computer science)	46.5
Jhanvi Patel	BSC(computer science)	44.5
Karishma Rao	BSC(nursing)	47.5
Krithika S	BSC(nursing)	39.5
Malavika Iyer	BSC(nursing)	47.5
Mohini Gupta	BSC(nursing)	43.5
Nanditha S	BSC(nursing)	46.5
Nithya Menon	BSC(nursing)	40.5
Pragathi N	BSC(augriculture)	45.5
Ragini Verma	BSC(augriculture)	47.5

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4.3 AI Tool Integration Results

- **Tool Used**: Microsoft Copilot was integrated with Microsoft Excel to assist in the analysis of the 561 student records.

- **Data Cleaning Assistance**: AI was used to identify inconsistencies, such as duplicate entries and formatting errors, which streamlined the preparation of the dataset.

- **Insight Generation**: Natural language prompts were used to identify key performance indicators (KPIs), such as the top-performing students and the average marks for each course.

- **Trend Analysis**: Copilot helped analyze the correlation between attendance and academic status, highlighting that low attendance was a significant factor in students receiving an "F" grade or "Fail" status.

- **Efficiency Gains**: The integration of AI tools significantly reduced the time required to generate summary reports and identify at-risk students compared to manual tracking methods.

- **Automated Summarization**: AI was utilized to draft professional summaries and findings based on the large volume of semester-wise data.

Sample AI Prompts Used

- "Identify students whose average marks are above 90% and list their respective courses."
- "Summarize the correlation between attendance percentages below 75% and the final pass/fail status."
- "Calculate the average marks for each course across all six semesters and identify the course with the lowest performance."

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

This section summarizes the primary insights discovered during the analysis of the 561 student records, such as the correlation between low attendance and failing grades.

5.2 Project Links

This section summarizes the primary insights discovered during the analysis of the 561 student records, such as the correlation between low attendance and failing grades.

5.3 Limitations

- **Static Data Source:** The project currently relies on static CSV/Excel exports, meaning the analysis does not update automatically when new semester marks are entered in a live database.

- **Manual Data Entry Errors:** Since the raw data was manually compiled, there is a risk of "Human Error" in the initial input phase which required extensive cleaning to rectify.

- **Limited Predictive Scope:** The current system uses descriptive analytics (what happened) but lacks predictive modeling to forecast exactly which students might fail in future semesters based on current trends.

- **Software Dependency:** The analysis and AI-assisted features are heavily dependent on a Microsoft 365 subscription and stable internet access for Copilot, which may not be available in all institutional environments.

- **Single Dimension Analysis:** While the project correlates attendance and marks, it does not account for external qualitative factors such as student health, socio-economic background, or teaching methodology changes.

5.4 Future Work

- **Integration of Real-Time Pipelines:** Future versions could use **Power Automate** to link the Excel sheet to a live Google Form or Microsoft Form for real-time data ingestion.

- **Predictive Analytics:** Implementation of machine learning algorithms (e.g., Linear Regression) to predict final CGPA based on the performance of the first two years.

- **Automated Alert System:** Developing a feature that automatically sends email notifications to students and parents when attendance drops below the 75% threshold.

- **Web-Based Dashboard:** Transitioning the Excel analysis into a cloud-based **Power BI service** dashboard accessible via mobile devices for faculty on the go.

- **Standardized Prompt Library:** Building a dedicated library of AI prompts specific to academic counseling to help faculty generate personalized feedback for students.

5.5 Conclusion

- **Objective Achievement:** This project successfully met all stated objectives, including the organization of academic records for 561 students and the establishment of a standardized grading system.

- **Actionable Insights:** The analysis proved that data-driven monitoring is superior to manual tracking, as it clearly identified students like **Ananya Rao** as "At-Risk" and recognized top performers like **Amtul Jaleel Ayesha**.

- **Impact of AI:** The integration of Microsoft Copilot demonstrated how AI can drastically reduce the time needed for complex data cleaning and insight generation, turning hours of work into minutes.

- **Institutional Value:** By correlating attendance with academic status, the project provides **Rishi UBR Women's College** with a blueprint for early intervention strategies that can improve overall pass rates.

- **Personal Growth:** The fellowship experience has significantly enhanced our practical skills in Data Analytics and AI-assisted reporting, preparing us for modern industry standards.

REFERENCES

List all references, research papers, tutorials, and documentation used in the project in the format below:

[1] Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

[2] Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/

[3] Microsoft Corporation. Microsoft Copilot — AI-Powered Productivity. Available at: https://copilot.microsoft.com

[4] Edunet Foundation. AICW Fellowship Program Guidelines. 2026.

[5] Appendix A — Sample Dataset | Student ID | Name | Course | Avg Marks | Grade | Status | Attendance% | | :--- | :--- | :--- | :--- | :--- | :--- | :---|

Key Formulas Used

To achieve these results, the following logic was applied:

- **Average Marks:** `=AVERAGE(E2:J2)` (Calculated across Semesters 1-6).
- **Grade Assignment:** `=IF(K2>=90,"A",IF(K2>=80,"B",...))` (Nested logic for performance classification).
- **Pass/Fail Status:** Logic determined by whether the average marks exceeded the passing threshold of 50.

APPENDICES

Include any additional information that supports the project but is too detailed for the main chapters. This may include:

Appendix A — Sample Dataset

Project STUDENT ANALYSIS.xlsx

Appendix B — Key Excel Formulas / DAX Measures Used

To achieve these results, the following logic was applied:

- **Average Marks:** `=AVERAGE(E2:J2)` (Calculated across Semesters 1-6).
- **Grade Assignment:** `=IF(K2>=90,"A",IF(K2>=80,"B",...))` (Nested logic for performance classification).
- **Pass/Fail Status:** Logic determined by whether the average marks exceeded the passing threshold of 50.

Appendix C — AI Prompts Used

- **Prompt 1:** "Analyze the attached student dataset of 561 records and identify the top 10 students based on their average marks across all six semesters."

- **Prompt 2:** "Identify students whose attendance is below 75% and determine if there is a direct correlation between their low attendance and a 'Fail' status in the 'Status' column."

- **Prompt 3:** "Calculate the average performance of students grouped by their course (e.g., B.com, BSC, BCA) and identify which course has the highest number of students with an 'A' grade."

- **Prompt 4:** "Identify any anomalies in the data, such as students with high average marks but extremely low attendance, and suggest potential reasons for these outliers."

- **Prompt 5:** "Generate a summary report for college management highlighting the percentage of students who passed versus those who failed, and provide three recommendations for improving the pass rate based on the attendance trends found."